The Treasure Map Mystery

Whodunnit? · Transformations Name: _____

At the Explorers' Society, members gathered for the big unveiling of the **Ancient Map**, which showed the way to a legendary lost island. But the glass case was bare!

Six people were in the building, each in a different place, and there are six ways to sneak a map out. You are the society detective: find **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Prof. Atlas Cartographer	☐ Map Room	☐ Rolled in a poster tube
☐ Ms. Compass Librarian	☐ Rare Books Vault	☐ Folded inside a book
☐ Explorer Rio Adventurer	☐ Globe Gallery	☐ Hidden in a globe
☐ Mr. Grid Surveyor	☐ Reading Nook	☐ Slipped in a backpack
☐ Captain Nova Pilot	☐ Courtyard Fountain	☐ Sent as a paper airplane
☐ Nora Webb Student Intern	☐ Rooftop Observatory	☐ Tucked in an umbrella

The culprit is _____ in _____ using _____

Show Your Work

The Treasure Map Mystery · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Treasure Map Mystery

Translate the point (2, 3) using the rule (x, y) → (x + 4, y − 1). Find the image.

A. (4, −1)	No paper airplane was found in the courtyard. The fountain stayed clear of paper.
B. (6, 4)	Explorer Rio was updating the world map in the Map Room. The curator saw his boots.
C. (6, 2)	Prof. Atlas was drawing coastlines in the Map Room, with students watching.
D. (−2, 4)	Prof. Atlas was tracing a compass rose on the Rooftop Observatory. A student held the lamp.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Treasure Map Mystery

Reflect the point (3, −2) across the x-axis. Find the image.

A. (−2, 3)	The paper airplane bin was empty. Nobody had folded one that day.
B. (−3, −2)	Explorer Rio was hunting for an old atlas in the Rare Books Vault. The vault log has his visit.
C. (3, 2)	Every poster tube was checked at the door. They held only posters.
D. (−3, 2)	Ms. Compass was reading star charts at the Rooftop Observatory. The telescope sign-in sheet shows her.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Treasure Map Mystery

Reflect the point (−4, 5) across the y-axis. Find the image.

A. (5, −4)	Every window was shut and locked. No airplane could have been sent out.
B. (4, 5)	Ms. Compass was shelving ancient books in the Rare Books Vault. The vault log shows her key card.
C. (4, −5)	Mr. Grid was surveying the Rooftop Observatory with a tripod. The tripod holes are still there.
D. (−4, −5)	Explorer Rio was napping in the Reading Nook with a travel book. A guard woke him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Treasure Map Mystery

Rotate the point (2, 3) 90° counterclockwise about the origin. Find the image.

A. (−3, 2)	The books were all flipped through. No map folded inside.
B. (3, −2)	Explorer Rio was resting by the Courtyard Fountain and eating a sandwich. A pigeon agrees.
C. (−2, −3)	Captain Nova was checking the weather at the Rooftop Observatory. Her notes are dated.
D. (3, 2)	The rooftop breeze was too weak to carry any paper plane. Nothing was flown.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Treasure Map Mystery

Rotate the point $(4, -1)$ 180° about the origin. Find the image.

A. $(-4, -1)$	Nora Webb was sweeping the Rooftop Observatory floor with a broom. The janitor confirms it.
B. $(-4, 1)$	Mr. Grid was measuring the Globe Gallery with a laser. The measuring log is time-stamped.
C. $(-1, -4)$	Not one paper airplane was found on the library grounds. Only leaves.
D. $(4, 1)$	Explorer Rio was unrolling maps in the Map Room all morning. The librarian counted them.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Treasure Map Mystery

Dilate the point $(3, -6)$ by a scale factor of $\frac{1}{3}$ with the center at the origin. Find the image.

A. $(1, -2)$	The big globe was opened. It held nothing but a dusty ruler.
B. $(1, 2)$	A search of the lawn and roof found no folded paper. No plane was launched.
C. $(9, -18)$	Explorer Rio was spinning the globes in the Globe Gallery. A tour group watched him.
D. $(-1, 2)$	Prof. Atlas was pointing at the stars from the Rooftop Observatory with the visiting class.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Treasure Map Mystery

Dilate the point (−2, 5) by a scale factor of 2 with the center at the origin. Find the image.

A. (4, −10)	The paper airplane contest was canceled that day. Nobody launched a single plane.
B. (−4, −10)	Mr. Grid was lining up a compass reading at the Rooftop Observatory. The logbook has his initials.
C. (0, 7)	Explorer Rio was borrowing a pen in the Reading Nook. Ms. Compass lent him one.
D. (−4, 10)	Captain Nova was reading in the Reading Nook. A guard checked on her twice.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Treasure Map Mystery

Reflect the point (1, 4) across the line y = x. Find the image.

A. (−4, −1)	Explorer Rio was dusting a shelf in the Rare Books Vault. The key card swipe backs this up.
B. (4, 1)	Every backpack was searched at the exit. No map inside.
C. (−1, 4)	The paper stack in the copy room was still sealed. No sheets were folded.
D. (1, −4)	Mr. Grid was leveling the telescope at the Rooftop Observatory. The level reading is time-stamped.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The translation (x, y) → (x − 3, y + 2) moves point A to A′. If A = (5, 1), find A′.

A. (8, 3)	The gardener swept the courtyard clean and found no paper plane. Not one.
B. (2, −1)	Captain Nova was setting up a sky camera on the Rooftop Observatory. The camera clock shows it.
C. (2, 3)	Nora Webb was feeding the koi at the Courtyard Fountain with a tour group.
D. (8, −1)	Explorer Rio was tossing coins at the Courtyard Fountain with a group of visitors.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

Rotate the point (−2, −5) 90° clockwise about the origin. Find the image.

A. (5, 2)	Ms. Compass was dusting the telescope in the Rooftop Observatory. Her cloth is still gray.
B. (5, −2)	Explorer Rio was tracing a trail on the big wall map in the Map Room. The guide took notes.
C. (−5, −2)	Every paper airplane in the lost-and-found was unfolded. All were plain homework.
D. (−5, 2)	The umbrella stand holds umbrellas only, with nothing tucked inside.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.